

Six Birds for Neutrino Mass: Lens-Swap Stability and Packaging Audits

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2 March 2026

Version 1

Abstract

Near-minimal cosmological bounds on the summed neutrino mass $\sum m_\nu$ can shift under seemingly benign pipeline/likelihood updates, yet standard presentations rarely include lens-level stability diagnostics that distinguish cosmology from packaging effects. We address the inconsistency highlighted by [1], where swapping the SPT-3G 2018 TT/TE/EE likelihood for the later SPT-3G D1 release—analyzing essentially the same sky field—tightens $\sum m_\nu$ constraints and can move posteriors toward the physical boundary $\sum m_\nu = 0$.

Our approach is to cast this lens swap as a packaging stability problem in the Six Birds / SBT formalism [2, 3]. We introduce a directional cross-lens audit functional, operationalize it via a profiled/localized/common-staging workflow, and demonstrate the resulting empirical stability picture under matched controls. Concretely, we (i) compute directional held-out mismatch statistics $\Delta\chi^2_{B|A}$ and $\Delta\chi^2_{A|B}$, (ii) localize mismatch by spectrum and multipole, (iii) repeat under a common-staging multipole-support control, and (iv) reproduce the $\sum m_\nu$ tightening under DESI DR2 BAO for two baselines (SPT-only+DESI; Planck-subset+PR4-lensing CMB baseline+DESI).

We find that cross-lens mismatch is large and asymmetric; profiling reduces but does not remove the failure; common-staging changes the profiled mismatch only at the few-percent level; and localization shows TT high- ℓ structure (roughly $\ell \sim 2000$ – 3000) dominates within the overlap region. Under fixed BAO, swapping SPT2018→D1 tightens the $\sum m_\nu$ 95% upper bound by $\mathcal{O}(10^{-1}, \text{eV})$ in the SPT-only+DESI regime and by $\mathcal{O}(10^{-2}, \text{eV})$ in the Planck-subset baseline+DESI regime.

These artifacts turn an ambiguous parameter shift into a reproducible, localized diagnosis of lens disagreement and provide a disciplined contract for follow-up tests (including raw-data reprocessing diagnostics already emphasized by [1]). We are explicit that identifying the pipeline root cause or establishing a definitive $\sum m_\nu$ value is not done here; our conclusions rely on audits of released likelihood objects under matched controls and the physical gate $\sum m_\nu \geq 0$.

keywords: neutrino mass; cosmological bounds; SPT-3G; lens stability; cross-lens audit; packaging mismatch; Six Birds framework

1 Introduction

The summed neutrino mass $\sum m_\nu$ is a primary target of cosmological inference, with constraints driven by the interplay between early-universe information from the CMB and late-time distance/growth information from large-scale structure. As limits approach the minimal-mass scale,

however, it becomes increasingly important to understand how much of an apparent “tightening” reflects cosmology versus how much reflects sensitivity to data processing, compression, nuisance parameterization, and dataset-combination route choices.

A sharp example is reported in [1]. They compare constraints obtained from two closely related SPT-3G likelihood releases: an SPT-3G 2018 TT/TE/EE baseline and a later 2019–2020 “D1” release. Holding fixed an otherwise matched baseline built from selected Planck 2018 (PR3) products and Planck PR4 lensing, they find that replacing SPT-3G 2018 with SPT D1 can tighten the $\sum m_\nu$ upper bound and shift the posterior toward the physical boundary $\sum m_\nu = 0$, which they describe as a “curious shift” given that both releases analyze essentially the same sky field but differ in observing seasons and analysis pipeline choices. They also emphasize that a decisive diagnostic would be to reprocess the 2018 raw data through the D1 pipeline; we explicitly acknowledge that this test is not our proposal and should not be treated as a novel idea here [1].

This note is intentionally narrow in scope: we do not attempt to identify the physical or instrumental root cause of the discrepancy, nor do we claim that any particular $\sum m_\nu$ bound is “the” correct answer. Instead, we operationalize the situation as a *lens-swap stability* problem and apply the Six Birds / SBT audit discipline [2, 3]. In SBT language, each likelihood release constitutes a distinct *lens* f (a data-processing and compression map into a likelihood object; not gravitational lensing), while a chosen macro-model class and nuisance parameterization define a *completion* U . The resulting packaged inference $E = U \circ f$ should be stable under benign lens variations if the package is coherent. When it is not, SBT recommends explicit audits and staging controls that localize where and how the mismatch enters.

What we do. We provide three concrete, falsifiable artifacts:

1. **Replication of the direction of the $\sum m_\nu$ shift under a lens swap.** Using a fully reproducible Cobaya-based pipeline, we reproduce the qualitative behavior that swapping SPT2018→D1 tightens $\sum m_\nu$ when combined with DESI DR2 BAO, both in an SPT-only+DESI setting and in a baseline that matches the Planck-subset+PR4-lensing structure of [1]. A compact summary is given in Table 4.
2. **Cross-lens audits with localization.** We quantify directional held-out mismatch between the two lenses using $\Delta\chi^2$ -style audit statistics and localize the mismatch by spectrum and multipole, producing a concrete “where to look” map rather than an abstract parameter-level tension.
3. **Common-staging controls and boundary-mode mechanism.** We explicitly control for multipole-support differences by enforcing a common staging, and we provide a minimal toy mechanism illustrating how boundary modes (“quasi-negative” behavior under an $\sum m_\nu \geq 0$ gate) can arise generically when inconsistent packaged constraints are combined and tightened.

A compact numerical summary of these shifts is given in Table 4.

Paper structure. Related literature is briefly summarized in Section 2. We restate the Six Birds / SBT primitives and map them to this setting in Section 3. Baseline definitions and implementation details follow in Sections 4 and 5. Audit and staging results are presented in Section 6, and the $\sum m_\nu$ shift results are presented in Section 7. We discuss boundary-mode interpretation in Section 8 and conclude with limitations and next diagnostics in Sections 9 and 10.

2 Related work

This note builds on the lens-swap observation in [1] and the Six Birds/SBT framing in [2, 3]. Our computations use standard cosmology inference and Boltzmann tools [4–6], DESI DR2 BAO likelihood data [7, 8], and Planck products [9, 10].

3 Six Birds / SBT framing

We summarize the minimal Six Birds / SBT primitives needed for this note and map them to the SPT2018→D1 lens swap [2, 3]. The role of this section is operational: it explains why a shift in a packaged parameter bound under a pipeline change should be treated as a first-class stability question, and why explicit audits and staging controls are the appropriate response.

Lenses, completions, and packaging. A *lens* is any data-processing and compression map that produces an inference object (here: a likelihood) from an underlying microstate. In this paper, a lens is emphatically *not* gravitational lensing; it is a pipeline/likelihood release. Let X denote the relevant microstate (cosmology plus nuisance/systematic degrees of freedom) and let

$$f : X \rightarrow \mathcal{L} \quad (3.1)$$

denote the map that takes the microstate to a likelihood object (or its sufficient summary). A *completion* specifies how we close the discarded information: a macro-model class, nuisance parameterization, and priors. In our setting, U is Λ CDM+ $\sum m_\nu$ with an explicit physical gate $\sum m_\nu \geq 0$ and the nuisance/foreground parameterization appropriate to each likelihood release. Packaging is the composition

$$E = U \circ f, \quad (3.2)$$

which maps the (unobserved) microstate to a packaged macro object such as a posterior distribution or a best-fit point. In the idealized limit of a coherent package, benign changes to the lens (e.g. changing observing season or pipeline implementation while viewing the same sky) should not drastically change the packaged macro output.

A directional audit statistic for lens swaps. SBT emphasizes that when two lenses disagree, one should quantify the mismatch *directionally* via train/test style audits [3]. Let $\log \mathcal{L}_A(\theta)$ and $\log \mathcal{L}_B(\theta)$ be the log-likelihoods for two lenses A and B under a shared parameter vector θ . Define the (possibly non-unique) best-fit estimator for each lens,

$$\hat{\theta}_L \in \arg \max_{\theta} \log \mathcal{L}_L(\theta), \quad L \in \{A, B\}. \quad (3.3)$$

The basic held-out mismatch statistic we use throughout is the directional improvement in B when moving from the A -fit to the B -fit:

$$\Delta \chi_{B|A}^2 \equiv -2 \left[\log \mathcal{L}_B(\hat{\theta}_A) - \log \mathcal{L}_B(\hat{\theta}_B) \right]. \quad (3.4)$$

By construction, $\Delta \chi_{B|A}^2 \geq 0$ when $\hat{\theta}_B$ is the maximizer found by our optimizer/sampler; large $\Delta \chi_{B|A}^2$ indicates that the A -packaged macro object predicts B poorly, even after accounting for B ’s own internal best-fit. Importantly, $\Delta \chi_{B|A}^2$ need not equal $\Delta \chi_{A|B}^2$: route mismatch is typically *directional*. When the likelihood supports blockwise decomposition (e.g. by spectrum and multipole band), $\Delta \chi_{B|A}^2$ can be localized as a sum of per-block contributions, turning an abstract mismatch into a concrete “where to look” diagnostic.

Table 1: Six Birds primitives (P1–P6) and their instantiation in the SPT2018↔D1 lens swap setting.

Primitive	SBT meaning (informal)	Instantiation in this work
P1 Rewrite	Minimal correction needed for macro closure when packaged dynamics fail to commute	We treat boundary sensitivity as a generic truncation effect under mismatch; <i>in a toy-only demonstration</i> , a one-mode surrogate rewrite reduces cross-lens $\Delta\chi^2$ without invoking new physics. We do not apply a rewrite-mode correction to the real SPT lens swap in this note.
P2 Gating	Constraints defining admissible macro objects and completions	Physical gate $\sum m_\nu \geq 0$; dataset inclusion/exclusion (e.g. no Planck high- ℓ TE/EE); nuisance parameterization choices.
P3 Route mismatch	Non-commutation / staging dependence of packaged outputs across routes	The lens swap $A = \text{SPT2018}$ vs $B = \text{SPT D1}$ yields different packaged inferences; we quantify route mismatch via directional $\Delta\chi_{B A}^2$ and $\Delta\chi_{A B}^2$.
P4 Staging	Dependence on scale, resolution, support, or cut choices	Common-staging control by matching multipole support across lenses; we test whether the mismatch is explained by ℓ -range differences.
P5 Packaging	The packaged macro object produced by $(U \circ f)$	Posteriors/best-fits for $\sum m_\nu$ and nuisance parameters are treated as packaged objects; instability under lens swap is a packaging-stability failure.
P6 Accounting / audit	Held-out prediction and decomposition of failures	Cross-lens audits and localization by spectrum and multipole provide an actionable diagnosis of where the packaged summaries disagree.

Staging and common-support controls. SBT distinguishes intrinsic lens mismatch from artifacts of *staging*—changes in effective resolution, smoothing, cuts, or support that can change the packaged object even under perfect internal consistency [2]. In CMB likelihoods, multipole cuts are a primary staging control. When comparing SPT2018 and SPT D1, we therefore implement a *common-staging* control in which the D1 likelihood is restricted (as much as feasible) to the SPT2018 multipole support, and we repeat the same directional audits. If mismatch persists under common-staging and remains localized inside the overlap region, the evidence points away from simple support/cut explanations.

Six Birds primitives (P1–P6) in this note. Table 1 summarizes the primitives and their instantiation here. The table is not meant to be exhaustive; it is a compact guide to the audit logic used in the remainder of the paper.

Mapping to the inconsistency. In the remainder of the paper, we treat the two SPT likelihood releases as two lenses on (nearly) the same sky. The question is not “which lens is correct,” but whether a packaged inference is *stable* under this lens change. When it is not, SBT prescribes two immediate steps: (i) quantify and localize mismatch via audits (Eq. (3.4)), and (ii) test whether staging controls eliminate the mismatch. Only after these checks does it become meaningful to interpret parameter-level shifts—including $\sum m_\nu$ tightening or boundary motion—as potentially cosmological rather than packaging-sensitive.

4 Data sets and baselines

Our goal is to isolate the effect of swapping one CMB *lens* for another while holding fixed, as much as feasible, the rest of the “baseline” structure emphasized in [1]. Throughout, we treat likelihood releases as lenses (pipeline likelihood objects), not as immutable ground truth. This section defines the two SPT lenses and the baseline dataset combinations used later for both audits and $\sum m_\nu$ inference.

4.1 SPT-3G lenses: 2018 vs D1

We work with two SPT-3G multifrequency TT/TE/EE likelihood releases, treated as distinct lenses and evaluated through `candl` [11]:

- **SPT2018 lens:** SPT-3G 2018 TT/TE/EE likelihood [12]
(Candl ID `candl_data.SPT3G_2018_TTTEEE_multifreq`).
The effective multipole ranges are TT $750 \leq \ell \leq 3000$ and TE/EE $300 \leq \ell \leq 3000$.
- **SPT D1 lens:** SPT-3G D1 (2019–2020) TT/TE/EE likelihood [13]
(Candl ID `spt_candl_data.SPT3G_D1_TnE_multifreq`).
The effective multipole ranges are TT $400 \leq \ell \leq 3000$ and TE/EE $400 \leq \ell \leq 4000$.

These definitions and ranges follow the baseline description in [1].

4.2 Planck subset and PR4 lensing baseline

To match the structure of the “CMB baseline” used in [1], we implement a Planck PR3 subset together with Planck PR4 lensing [14]:

- **Low- ℓ Planck PR3:** TT for $\ell < 30$ and EE for $\ell < 30$ (we use the `EE_sroll2` variant available in Cobaya’s Planck likelihood family).
- **Truncated high- ℓ Planck PR3 TT:** TT only for $30 \leq \ell \leq 756$ via `planck_2018_highl_plik.TT_lite_native` with `bins_for_L_range=[30,756]`.
- **No Planck high- ℓ TE/EE:** in accordance with [1], we do not include high- ℓ Planck TE/EE likelihoods.
- **Planck PR4 lensing:** we include the public PR4 (NPIPE) lensing likelihood [10, 15] via `planckpr4lensing.PlanckPR4Lensing`.

We emphasize that this is an operational baseline intended to match the selection described in [1]; it is not the full Planck 2018 likelihood [9].

A compact summary of the two CMB baselines used throughout is given in Table 2.

Implementation note: our runs use the released SPT TT/TE/EE likelihood objects for SPT2018 and D1, together with the Planck subset and Planck PR4 lensing. We do not include a separate SPT lensing-reconstruction likelihood component. Therefore, our “Planck-subset baseline” should be read as matching the Planck selection in [1] while only partially matching their CMB baseline composition; this is a plausible contributor to residual numerical differences in $\sum m_\nu$ bounds.

Table 2: Baseline definitions as described in [1] (CMB and CMB-D1). In our implementation we match the Planck-subset + PR4 lensing structure and swap only the SPT TT/TE/EE likelihood release; see text for any deviations.

Baseline	SPT primary (ℓ ranges)	Planck PR3 primary subset	Lensing
CMB (SPT2018)	TT $[\ell_{\min}, \ell_{\max}] = [750, 3000]$; TE/EE $[\ell_{\min}, \ell_{\max}] = [300, 3000]$	TT $\ell < 30$ (Commander); EE $\ell < 30$ (SRoll2); TT $30 < \ell < 756$ (Plik).	SPT lensing + Planck PR4 lensing
CMB-D1	TT $[\ell_{\min}, \ell_{\max}] = [400, 3000]$; TE/EE $[\ell_{\min}, \ell_{\max}] = [400, 4000]$	TT $\ell < 30$ (Commander); EE $\ell < 30$ (SRoll2); TT $30 < \ell < 756$ (Plik).	SPT lensing + Planck PR4 lensing

4.3 DESI DR2 BAO likelihood

For BAO, we implement a Gaussian likelihood for DESI DR2 BAO using the publicly documented DR2 BAO mean vectors and covariance matrices [7, 8, 16, 17]. The implemented observable vector includes (D_M/r_d) , (D_H/r_d) , and (D_V/r_d) points from the DR2 Gaussian BAO compression for the “ALL” combined set [7, 8, 18]. This combined set includes low-redshift galaxy BAO and Ly α forest BAO points, so our “ALL” vector is not purely galaxy BAO. Theory predictions are computed from the same cosmological backend used for CMB inference (Section 5.4).

4.4 Non-novelty note: pipeline reprocessing diagnostic

The baseline-swap sensitivity highlighted in [1] naturally motivates a decisive diagnostic: reprocessing the 2018 raw data through the D1 pipeline to separate sky information from pipeline effects. We explicitly acknowledge that this diagnostic is already emphasized in [1]; we do not claim novelty for proposing it. Our focus here is complementary: we provide a packaging/audit framework and a set of concrete, localizable metrics that quantify and stage-control the lens mismatch in the packaged likelihood objects available today.

5 Methods

This section summarizes (i) the audit metrics used to quantify and localize cross-lens mismatch, (ii) the common-staging control, and (iii) the cosmological inference setup used to estimate $\sum m_\nu$ posteriors in the presence of lens swaps.

5.1 Directional cross-lens audits and localization

Our primary audit statistic is the directional held-out mismatch $\Delta\chi^2_{B|A}$ defined in Eq. (3.4). Intuitively, $\Delta\chi^2_{B|A}$ asks: if we package the macro object using lens A , how poorly does that packaged solution predict lens B relative to B ’s own internal best fit? Because route mismatch is directional, we report both $\Delta\chi^2_{B|A}$ and $\Delta\chi^2_{A|B}$.

When the likelihood supports blockwise decomposition (e.g. by spectrum and multipole band), we additionally localize mismatch by writing $\Delta\chi^2_{B|A}$ as a sum over blocks. In our implementation, blocks correspond to a spectrum type (TT/TE/EE) and an ℓ -bin group. The result is a heatmap-style diagnostic that turns a global mismatch into a concrete localization: which spectrum and which multipole ranges dominate the held-out failure.

Blockwise accounting. When a likelihood provides additive blockwise contributions (e.g. by spectrum and multipole groups), we write

$$\log \mathcal{L}_B(\theta) = \sum_{g \in \mathcal{G}} \log \mathcal{L}_{B,g}(\theta), \quad (5.1)$$

and define the blockwise held-out mismatch contributions

$$\Delta\chi_{B|A,g}^2 \equiv -2 \left[\log \mathcal{L}_{B,g}(\hat{\theta}_A) - \log \mathcal{L}_{B,g}(\hat{\theta}_B) \right]. \quad (5.2)$$

Under Eq. (5.1) these satisfy the sum rule $\Delta\chi_{B|A}^2 = \sum_{g \in \mathcal{G}} \Delta\chi_{B|A,g}^2$ (and similarly for $A|B$), so the localization heatmaps should be read as an accounting decomposition of the same directional audit statistic rather than an unrelated diagnostic. We emphasize that $\Delta\chi^2$ is reported in the native $-2\Delta \log \mathcal{L}$ units of the packaged likelihood objects and is used here as a comparative diagnostic (across directions and staging controls), not as a calibrated hypothesis-test p-value; null calibration would require additional held-out or posterior-predictive constructions beyond the scope of this short note.

5.2 Profiled audits: separating shared parameters from nuisance packaging

A key subtlety is that lenses can differ in nuisance parameterization and calibration degrees of freedom. To more closely align the audit with SBT’s “completion” semantics, we also implement a *profiled* variant of the cross-lens audit. For direction $B|A$, we:

1. hold fixed a small set of *shared* parameters (the intersection of parameters we choose to treat as common across lenses), and
2. re-optimize a tractable subset of B ’s nuisance parameters (a calibration/bandpower-mode subset) to maximize $\log \mathcal{L}_B$ under those shared parameters.

We then compute a profiled $\Delta\chi_{B|A}^2$ by comparing B ’s likelihood at the profiled A -shared point against B ’s own profiled best fit. Operationally, this separates (i) mismatch that can be absorbed by lens-specific nuisance packaging from (ii) mismatch that persists even after giving the test lens freedom to retune its nuisance degrees of freedom.

In practice, full profiling over all nuisance parameters can be computationally expensive. We therefore use a runtime-capped profiling set consisting of a calibration/beta subset (referred to internally as `TT_cal_beta`), reduced to a tractable number of parameters for deterministic optimization.

Formal definition. Let parameters be partitioned as $\theta = (\phi, \eta_B)$ where ϕ denotes the shared parameters we hold fixed in the audit direction $B|A$ and η_B denotes nuisance parameters specific to lens B . Define the profiled log-likelihood

$$\log \tilde{\mathcal{L}}_B(\phi) \equiv \max_{\eta_B} \log \mathcal{L}_B(\phi, \eta_B), \quad (5.3)$$

and the profiled directional mismatch

$$\widetilde{\Delta\chi}_{B|A}^2 \equiv -2 \left[\log \tilde{\mathcal{L}}_B(\hat{\phi}_A) - \log \tilde{\mathcal{L}}_B(\hat{\phi}_B) \right], \quad (5.4)$$

where $\hat{\phi}_L$ is the shared-parameter component of the best-fit point under lens L . In practice we approximate Eq. (5.3) by profiling only a tractable subset of B ’s nuisance parameters under a runtime cap; the intent is not to “profile away” all mismatch, but to reduce sensitivity to purely lens-specific packaging degrees of freedom. In our implementation, the profiled set comprises the following calibration/beta parameters:

Tcal_ext150, Tcal_rel90, Tcal_rel220, Ecal_ext150, Ecal_rel90, Ecal_rel220,
 TT_CIBClustering_Alpha, TT_CIB_150x150, TT_CIB_150x220, TT_CIB_220x220,
 TT_GalCirrus_Alpha, TT_GalCirrus_Amp.

For the reverse direction, the capped set is:

TT_tSZ_CIB_Corr_Amp, TT_tSZ_CIB_corr, Ecal150, Ecal220, Ecal90,
 TT_CIBClustering_Alpha, TT_CIBClustering_Amp, TT_CIBClustering_Beta,
 TT_GalCirrus_Alpha, TT_GalCirrus_Amp, TT_GalCirrus_Beta, TT_Poisson_150x150

(runtime-capped to 12 parameters; see run metadata).

5.3 Common-staging control

Because the two SPT lenses differ in multipole support, a basic staging control is to restrict the broader-support lens to the narrower-support range and repeat the same audits. We implement a *common-staging* control in which the D1 lens is restricted (as much as feasible) to the SPT2018 multipole support (TT: $\ell \geq 750$; TE/EE: $\ell \geq 300$; all with $\ell \leq 3000$). The restriction is implemented at likelihood-instantiation time via selection masks recorded in the run metadata. If mismatch persists under common-staging and remains localized inside the overlap region, this points away from a pure support/cut explanation. This common-staging control matches *multipole support* but does not fully equalize the compressed summaries: the two releases retain their own bandpower binning/window functions and internal covariance conventions, so this test should be interpreted as a support/cut control rather than an identity-of-data-vector comparison.

5.4 $\sum m_\nu$ inference pipeline

We estimate $\sum m_\nu$ posteriors using Cobaya [4] with a standard Boltzmann backend (CLASS [5] and/or CAMB [6], depending on the configuration). The canonical SPT-only+DESI results use **CAMB** while the Planck-subset baseline+DESI results use **CLASS**; see run metadata for exact configs. We sample the standard six Λ CDM parameters $\{\omega_b h^2, \omega_c h^2, H_0, \ln(10^{10} A_s), n_s, \tau\}$ together with $\sum m_\nu$ under a physical gate $\sum m_\nu \geq 0$. Following the neutrino modeling description in [1], our Λ CDM+ $\sum m_\nu$ completion assumes three degenerate massive neutrino states when $\sum m_\nu$ is allowed to vary.

Implementation caveat (CLASS parameterization). CLASS expects massive neutrinos to be specified via N_{ncdm} and per-species masses m_{ncdm} . To keep $\sum m_\nu$ explicitly present in the recorded chains and summaries while providing CLASS the required per-species masses, we sample an internal parameter (denoted `mnu_sample`) and define a derived parameter $\sum m_\nu$ together with $m_{\text{ncdm}} = (\sum m_\nu/3, \sum m_\nu/3, \sum m_\nu/3)$. This is a parameterization workaround; it preserves the reported $\sum m_\nu$ posterior but should be kept in mind when comparing config syntax to other pipelines.

Priors and stability checks. The lower bound $\sum m_\nu \geq 0$ is explicit in [1]. The upper bound of the prior is not stated in the paper source we audited; for reproducibility we therefore use a broad flat prior with upper bound 5 eV and report this choice explicitly. Since the resulting posteriors and 95% upper bounds we report are all $\ll 1$ eV, this upper bound is not active in practice; the relevant gate for boundary behavior is the physical constraint $\sum m_\nu \geq 0$. For each production run we record basic chain diagnostics (split- $\hat{R}$ [19, 20] and a crude ESS estimate [20] for $\sum m_\nu$) and a half-chain stability proxy for the 95% upper bound. We also perform a mild stability check by repeating (or partially re-running) one production configuration with a distinct random seed and

verifying that the 95% upper bound does not change materially. Posterior summaries (quantiles, KDE-based 1D densities, and simple convergence heuristics) are computed using GetDist [21].

5.5 Reproducibility note

All results in this note are tied to canonical run bundles recorded in `docs/findings/freeze_checklist.md` and summarized in `docs/findings/canonical_results.json`. All figures and tables appearing in the paper are curated from, or auto-generated against, canonical run bundles and a canonical results summary, so the numerical claims in the text are backed by a deterministic artifact pipeline rather than manual transcription. The full analysis code, experiment runners, and manuscript build workflow are available at <https://github.com/ioannist/six-birds-neutrino>. The figures curated under `paper/figures/` and the tables under `paper/tables/` are generated from these canonical artifacts. For convenience, the primary run bundles cited later include:

- SPT-only + DESI $\sum m_\nu$ shift comparison: `runs/20260212_080120_648405_mnu_shift_spt_only_desi`.
- Planck-subset baseline + DESI $\sum m_\nu$ shift comparison: `runs/20260212_111357_576743_mnu_shift_cmb_plancksubset_desi`.
- Profiled baseline vs common-staging audit comparison: `runs/20260211_181258_958327_spt_profiled_common_staging_compare`.

A one-command rebuild of key artifacts is provided by `make freeze_check`, and the neutrino inference workflows are exposed via `make mnu_smoke` and `make mnu_full` when required external assets are present.

6 Audit results: lens mismatch, localization, and staging

We now treat the SPT2018 likelihood and the SPT D1 likelihood as two *lenses* on (nearly) the same sky field. The central question is whether packaged outputs are stable under swapping these lenses. Following Eq. (3.4), we quantify stability via directional held-out mismatch statistics and then localize the mismatch by spectrum and multipole.

6.1 Directional mismatch is large and asymmetric

Table 3 summarizes the directional cross-lens audit statistics for both the *unprofiled* audit (evaluating each lens at the other lens’s best-fit point) and the *profiled* audit (re-optimizing a tractable subset of the test lens’s nuisance parameters; Section 5.2). Two qualitative features are immediate:

1. **Mismatch is directional.** The values of $\Delta\chi^2_{B|A}$ and $\Delta\chi^2_{A|B}$ differ substantially, indicating that route mismatch is not symmetric across the two lenses.
2. **Mismatch is large even after profiling.** Profiling reduces sensitivity to purely lens-specific nuisance packaging, but does not eliminate the held-out failure, indicating that the mismatch is not solely a matter of nuisance retuning.

Table 3: Directional cross-lens audit summary for SPT2018 vs SPT D1, with and without common-staging. The unprofiled audit uses direct evaluation of the test lens at the train lens’s best-fit point; the profiled audit re-optimizes a tractable subset of test-lens nuisance parameters at fixed shared parameters (Section 5.2). “Common-staging” restricts the D1 lens to the SPT2018 multipole support (Section 5.3).

Audit	Direction	Baseline $\Delta\chi^2$	Common-staging $\Delta\chi^2$	Reduction (%)
Unprofiled	D1 2018	2222.17	2222.18	-0.00
Unprofiled	2018 D1	313.57	315.06	-0.48
Profiled	D1 2018	859.64	840.19	2.26
Profiled	2018 D1	369.52	370.46	-0.25

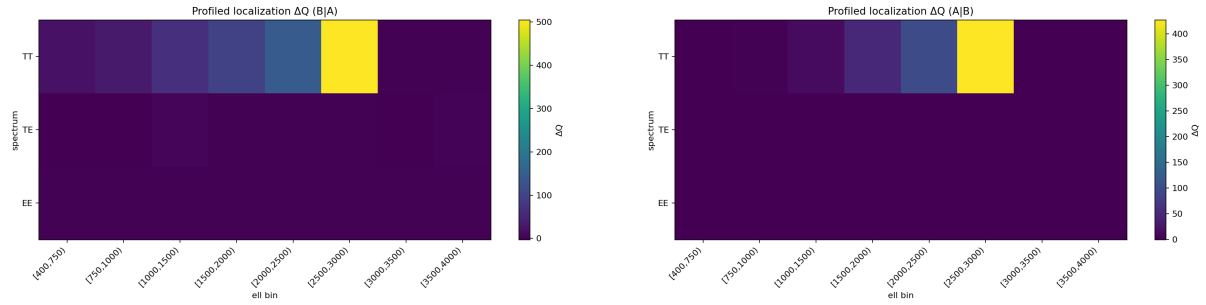


Figure 1: Profiled localization heatmaps for the SPT lens swap, decomposing the directional held-out mismatch into contributions grouped by spectrum type and multipole ranges. **Left:** $\Delta\chi^2_{B|A}$ (SPT D1 given SPT2018). **Right:** $\Delta\chi^2_{A|B}$ (SPT2018 given SPT D1). Darker regions indicate larger contributions to the held-out failure. In both directions, the dominant contributions arise from TT at high multipoles within the overlap region.

6.2 Localization points to TT high- ℓ structure in the overlap region

To make the mismatch actionable, we localize $\Delta\chi^2$ by decomposing it into contributions grouped by spectrum type and multipole ranges. Figure 1 shows the resulting localization heatmaps for the *profiled* audit in both directions. The dominant contributions arise from TT at high multipoles (notably the $\ell \sim 2000$ – 3000 range), while TE/EE contributions are comparatively small. Because high- ℓ TT is foreground- and calibration-sensitive, this localization should be read as identifying where the *packaged likelihood objects* disagree (e.g., foreground modeling, calibration/beam/transfer-function conventions, or covariance structure), not as a map-level diagnosis of the sky.

A key observation is that these multipoles lie well *inside* the overlap region of the two SPT lenses. In other words, the mismatch is not primarily driven by the fact that D1 extends to lower ℓ in TT or to higher ℓ in TE/EE. Rather, the held-out failure is concentrated in TT structure that both lenses nominally cover. This localization supports interpreting the “curious shift” highlighted in [1] as a packaging stability failure tied to how the two lenses compress and model the same underlying sky information.

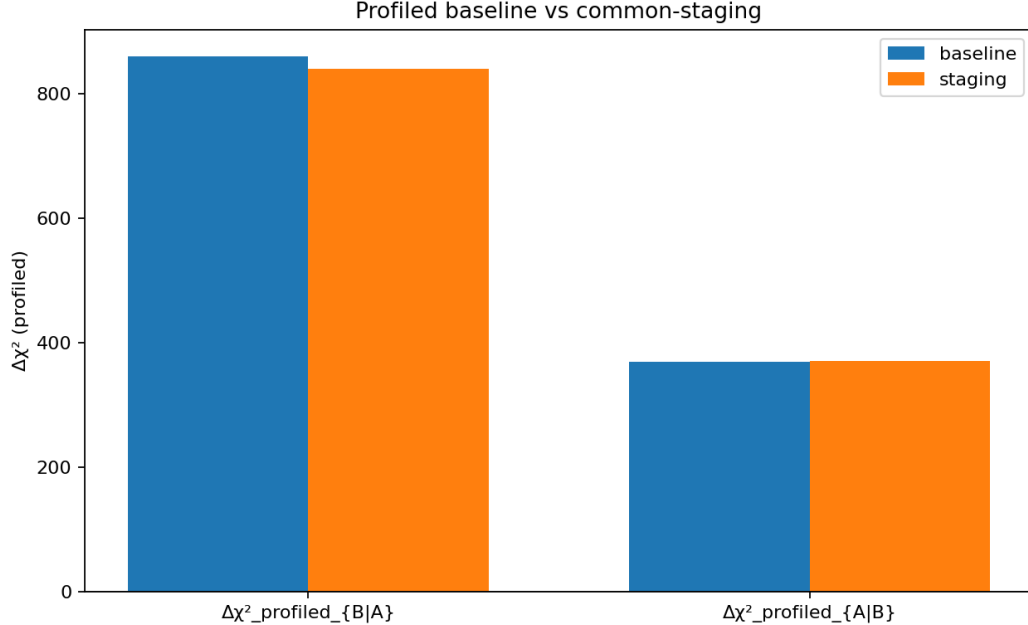


Figure 2: Common-staging control for the profiled cross-lens audit. Baseline uses the full D1 multipole support; common-staging restricts D1 to the SPT2018 support. The change in $\Delta\chi^2$ is small compared to the overall mismatch, indicating that staging alone does not explain the held-out failures.

6.3 Common-staging changes mismatch only at the few-percent level

A natural hypothesis is that the mismatch is driven by differences in multipole support between SPT2018 and D1. To test this, we implement a common-staging control in which the D1 lens is restricted (as much as feasible) to the SPT2018 multipole support and we repeat the same audits (Section 5.3).

Figure 2 visualizes the baseline-vs-common-staging comparison for the *profiled* audit. The resulting changes are modest: common-staging does not qualitatively resolve the mismatch, and the localization remains dominated by TT high- ℓ structure within the overlap region. This control therefore argues against a purely support/cut explanation for the lens mismatch and motivates treating parameter-level shifts (including $\sum m_\nu$ tightening) as packaging-sensitive unless accompanied by explicit lens-swap audits.

7 Results: $\sum m_\nu$ shift under a lens swap

We now report the $\sum m_\nu$ -posterior shifts induced by swapping the SPT lens (SPT2018 $\rightarrow$ SPT D1) under a fixed completion (Section 5.4). We emphasize the intended interpretation: these shifts are *packaged* outputs and should be read alongside the lens-swap audits in Section 6, which localize where the two lenses disagree and test whether staging controls resolve the mismatch.

7.1 SPT-only + DESI DR2 BAO: a large tightening under D1

We first consider an intentionally simplified regime: the SPT TT/TE/EE likelihood alone (SPT2018 or D1) combined with DESI DR2 BAO. This setting is useful because it isolates the effect of

Table 4: Summary of $\sum m_\nu$ shifts reproduced in this note. Values are in eV; “Shift” denotes (SPT2018 minus SPT D1) for the corresponding setup.

Setup	Lens	median [eV]	95% upper [eV]	mode [eV]	boundary frac
<i>SPT-only + DESI DR2 BAO</i>					
	SPT2018	0.071	0.194	0.050	0.00223
	SPT D1	0.042	0.085	0.070	0
	Shift (2018 – D1)	0.029	0.109	—	—
<i>Planck-subset baseline + PR4 lensing + SPT + DESI DR2 BAO</i>					
	SPT2018 baseline	0.068	0.080	0.075	0
	SPT D1 baseline	0.054	0.068	0.054	0
	Shift (2018 – D1)	0.013	0.012	—	—

swapping the SPT lens while holding fixed the late-time distance information. As summarized in Table 4, the D1 lens yields a substantially tighter $\sum m_\nu$ upper bound than the 2018 lens in this regime.

Figure 3 overlays the $\sum m_\nu$ posteriors for this SPT-only+DESI comparison. The qualitative effect is robust: under the same BAO likelihood and completion class, swapping SPT2018→D1 tightens $\sum m_\nu$.

7.2 Planck-subset baseline + PR4 lensing + DESI: headline replication with a smaller shift

We next run a baseline designed to mirror the Planck-subset+PR4-lensing structure described in [1] (Section 4.2), and we again combine with DESI DR2 BAO. In this setting the *direction* of the lens-swap effect matches the headline behavior emphasized in [1]: swapping SPT2018→D1 tightens the $\sum m_\nu$ upper bound. However, the magnitude of the tightening in our reproduction is smaller than the difference between the corresponding values quoted in [1] (Table 4). Because this baseline shift is at the $\mathcal{O}(10^{-2}, \text{eV})$ level, we treat it as a *directional* replication: its magnitude is comparable to our seed-based stability proxy for the 95% upper bound, so we avoid over-interpreting small numerical differences.

Figure 4 shows the $\sum m_\nu$ posterior overlay for this baseline+DESI setting together with a compact bounds summary table. In these production runs we do not observe strong posterior piling at the physical boundary $\sum m_\nu = 0$ (the boundary-fraction diagnostic is near zero), but a boundary mode can arise generically under truncation when inconsistent packaged constraints are tightened; we return to this mechanism in Section 8.

7.3 Comparison to [1] and likely drivers of residual differences

For context, [1] reports 95% upper bounds of $\sum m_\nu < 0.11 \text{ eV}$ for their SPT2018-based CMB baseline + DESI DR2 BAO, and $\sum m_\nu < 0.069 \text{ eV}$ for the corresponding CMB-D1 baseline + DESI DR2 BAO. Our reproduction matches the *direction* highlighted in [1]: under DESI DR2 BAO, swapping SPT2018→D1 tightens the $\sum m_\nu$ upper bound. The *magnitude* of the shift depends on details of the completion and likelihood packaging. Several plausible contributors to the remaining numerical differences (particularly in the SPT2018 baseline bound) include:

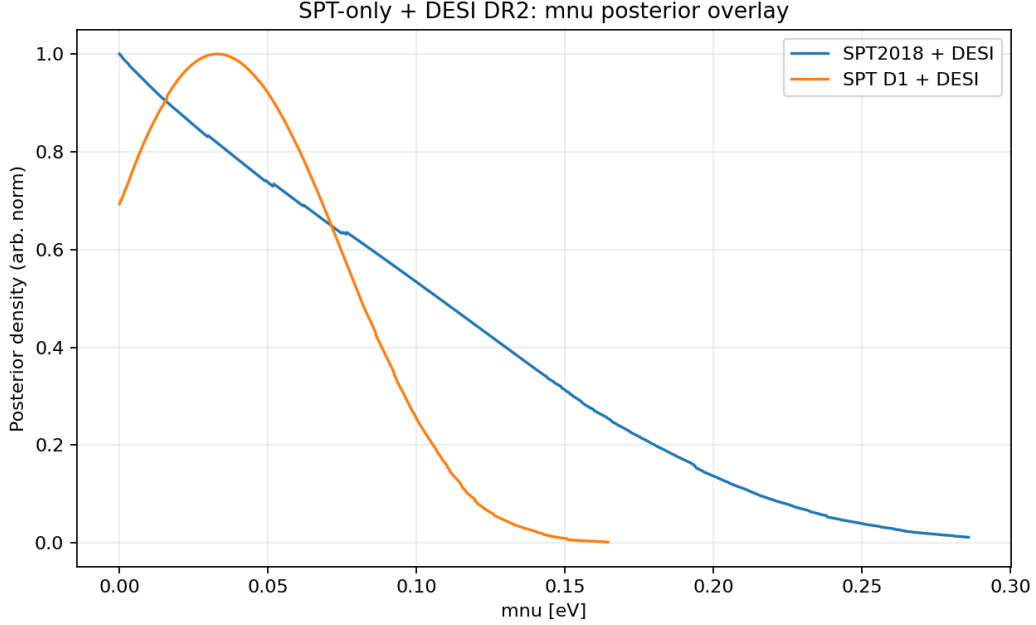
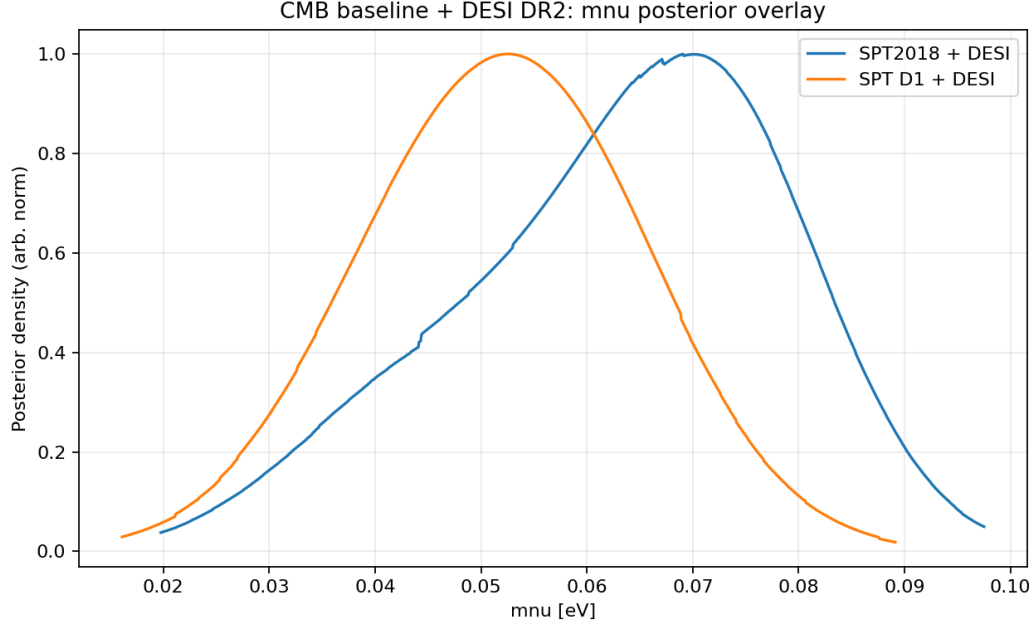


Figure 3: $\sum m_\nu$ posterior overlay for SPT-only + DESI DR2 BAO. The D1 lens yields a tighter $\sum m_\nu$ constraint than the SPT2018 lens under the same BAO likelihood and completion. Numerical summaries are given in Table 4.

- **Nuisance freedom and priors.** We use a deliberately minimal nuisance sampling set for computational tractability in production chains; tighter or looser bounds can result when additional nuisance parameters are varied, reparameterized, or assigned different priors.
- **Planck low- ℓ EE variant.** Our Planck subset uses the `EE_sro112` low- ℓ EE likelihood variant available in Cobaya; small baseline differences can propagate into tight derived bounds.
- **Likelihood and pipeline versions.** Even when dataset selections are conceptually matched, differences in released likelihood implementations, calibration conventions, and default settings can shift posteriors in the sub-0.1 eV regime.
- **Baseline components.** Our baseline matches the Planck-subset and PR4 lensing structure described in [1], but exact numerical agreement can depend on whether additional baseline components (and their nuisance couplings) are included and how they are parameterized in a given code stack.
- **SPT lensing component.** The CMB baseline in [1] includes an SPT lensing-reconstruction likelihood component; our baseline implementation does not, which can affect absolute tight bounds and may reduce the shift magnitude.

These sensitivities are precisely why we advocate pairing headline parameter bounds with explicit lens-swap audits and staging controls (Section 6): when the packaged likelihood objects disagree in localized regions of the summary statistics, parameter-level shifts should be interpreted as packaging-sensitive until the mismatch is resolved.



run	median	p95_upper	mode	boundary_fraction
SPT2018 + Planck subset + DESI	0.067767	0.079814	0.074696	0.000000
SPT D1 + Planck subset + DESI	0.054342	0.067653	0.054432	0.000000

Figure 4: $\sum m_\nu$ posterior overlay (top) and bounds summary (bottom) for the Planck-subset baseline + PR4 lensing + SPT + DESI DR2 BAO setting. The D1 lens again yields a tighter $\sum m_\nu$ constraint than SPT2018, but the shift magnitude is smaller than the gap reported in [1]. See Table 4 for the numerical comparison.

7.4 Implementation caveat: sampled `mnu_sample` and derived $\sum m_\nu$

As noted in Section 5.4, our Planck-subset baseline production runs are CLASS-based and parameterize massive neutrinos via three degenerate species masses $m_{\text{ncdm}} = (\sum m_\nu/3, \sum m_\nu/3, \sum m_\nu/3)$. To keep $\sum m_\nu$ explicit in chain outputs while providing CLASS the required per-species masses, we sample an internal parameter `mnu_sample` and expose $\sum m_\nu$ as a derived parameter used for reporting and plotting. This is a bookkeeping choice for tool compatibility; it does not change the intended meaning of the reported $\sum m_\nu$ posterior.

8 Boundary modes as a generic truncation mechanism

A recurring qualitative feature in near-minimal $\sum m_\nu$ inference is posterior behavior near the physical gate $\sum m_\nu \geq 0$. In particular, [1] describes cases in which swapping the SPT lens (SPT2018 $\rightarrow$ D1) and combining with external data can push the posterior toward the prior border at $\sum m_\nu = 0$. Here we provide a deliberately minimal mechanism showing that such boundary behavior can arise

generically from (i) combining incompatible packaged constraints and (ii) applying a hard truncation, without invoking negative masses as a physical possibility.

8.1 Two constraints plus a gate: mode at the boundary when the unconstrained optimum is negative

Consider a single nonnegative parameter $m \geq 0$ (standing in for $\sum m_\nu$) and suppose two independent packaged constraints yield approximately Gaussian log-likelihood contributions,

$$\log \mathcal{L}_i(m) = -\frac{(m - \mu_i)^2}{2\sigma_i^2} + \text{const}, \quad i \in \{1, 2\}, \quad (8.1)$$

with $\sigma_i > 0$. The combined (unconstrained) likelihood is proportional to a Gaussian with mean and variance

$$\mu_\star = \frac{\mu_1/\sigma_1^2 + \mu_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}, \quad \sigma_\star^2 = \frac{1}{1/\sigma_1^2 + 1/\sigma_2^2}. \quad (8.2)$$

If the two packaged constraints are in tension, it is possible for $\mu_\star < 0$ even though the physical parameter space is $m \geq 0$. Enforcing the gate $m \geq 0$ corresponds to truncating the Gaussian at 0, producing a (normalized) truncated density on $[0, \infty)$. In that case the mode occurs at

$$m_{\text{mode}} = \max(0, \mu_\star), \quad (8.3)$$

so whenever the unconstrained optimum lies at $\mu_\star \leq 0$, the *constrained* mode is exactly at the boundary $m = 0$. This is the sense in which “boundary modes” (sometimes informally described as “quasi-negative” behavior) can arise: the negative optimum is not physical, but it is the optimizer’s reconciliation point for incompatible packaged constraints, and truncation forces the packaged macro object to pile up at the boundary.

Figure 5 illustrates one such case: the unconstrained combined density peaks at a negative $\mu_\star$, while the truncated density on $m \geq 0$ peaks at $m = 0$.

Optional formal check (Lean). For completeness, we mechanically verified the key monotonicity statement underlying Eq. (8.3) in Lean4 [22, 23]: for $\sigma > 0$ and $\mu \leq 0$, the unnormalized density $x \mapsto \exp(-((x - \mu)^2/(2\sigma^2)))$ is decreasing on $[0, \infty)$ and hence achieves its maximum at $x = 0$. The proof is included in the repository.¹ This is not essential to the cosmological argument, but it documents the boundary-mode claim as a simple, checkable mathematical fact.

8.2 Tightening increases boundary-mode frequency

The practical relevance is that as constraints tighten (smaller σ_i), modest shifts in the effective means μ_i —including shifts induced by lens mismatch—can more readily push $\mu_\star$ below 0. We therefore performed a small sweep in which one constraint is held fixed while the other is tightened, and we estimated the fraction of sweep points for which the truncated posterior mode lies at the boundary. Figure 6 shows that the boundary-mode frequency can rise sharply as one constraint tightens.

¹See `lean/trunc_gauss_proof`; lemmas `TruncGaussProof.unnorm_antitoneOn_Ici` and `TruncGaussProof.unnorm_le_at_zero`.

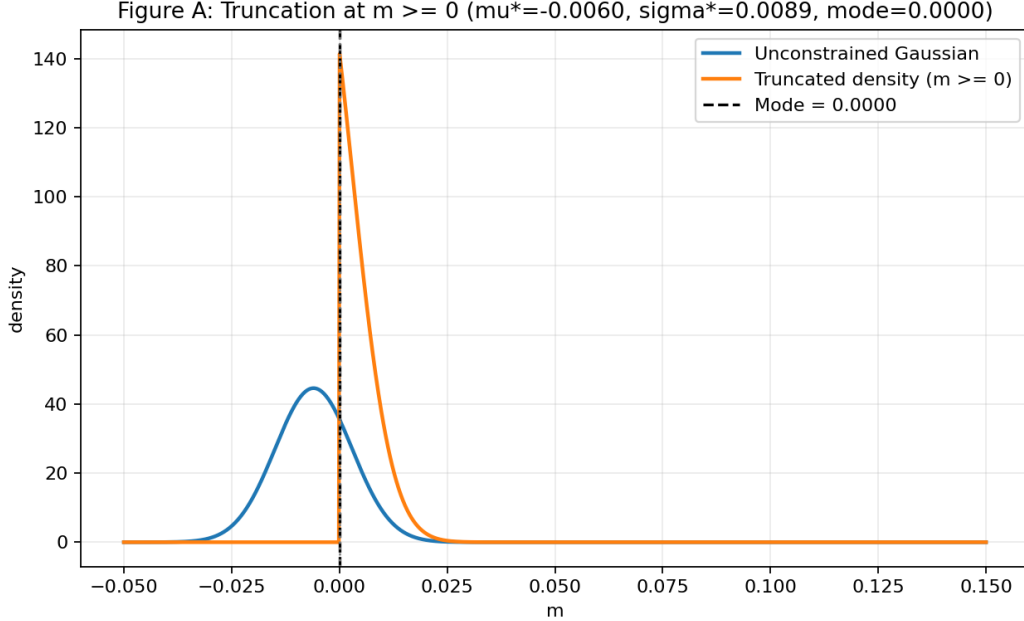


Figure 5: Toy mechanism: combining two Gaussian constraints can yield an unconstrained optimum at $\mu_\star < 0$. Enforcing a physical gate $m \geq 0$ truncates the density and forces the mode to the boundary at $m = 0$. This illustrates a generic mechanism for boundary modes without interpreting negative masses as physical.

8.3 Interpretation for $\sum m_\nu$ under lens swaps

This toy model does not claim that any specific real-data run must exhibit a boundary mode, and indeed our baseline+DESI production runs have near-zero boundary fraction for $\sum m_\nu$ (Section 7). Rather, it clarifies why boundary motion can be a *generic symptom* of packaging instability when a hard physical gate is present. In the SBT framing, the boundary is a gating constraint (P2), and strong boundary sensitivity is naturally interpreted as an indicator that the completion is being asked to reconcile incompatible packaged summaries (a route mismatch symptom, P3), especially when the mismatch is localized in the overlap region (Section 6).

Therefore, when [1] observes that swapping SPT2018→D1 can push $\sum m_\nu$ toward the $\sum m_\nu = 0$ prior border in some dataset combinations, a conservative reading is: the packaged constraints are sufficiently tight and sufficiently route-sensitive that small lens differences can move the unconstrained optimum across the gate. This strengthens (rather than weakens) the case for reporting explicit lens-swap audits and staging controls alongside headline bounds.

9 Discussion, limitations, and next diagnostics

9.1 What we learn from lens-swap auditing

The inconsistency emphasized by [1] is not merely a parameter-level curiosity; it is a stability question about packaged inference under a lens swap. In the Six Birds / SBT framing [2, 3], the two SPT likelihood releases define two lenses, and the posterior for $\sum m_\nu$ is a packaged macro object

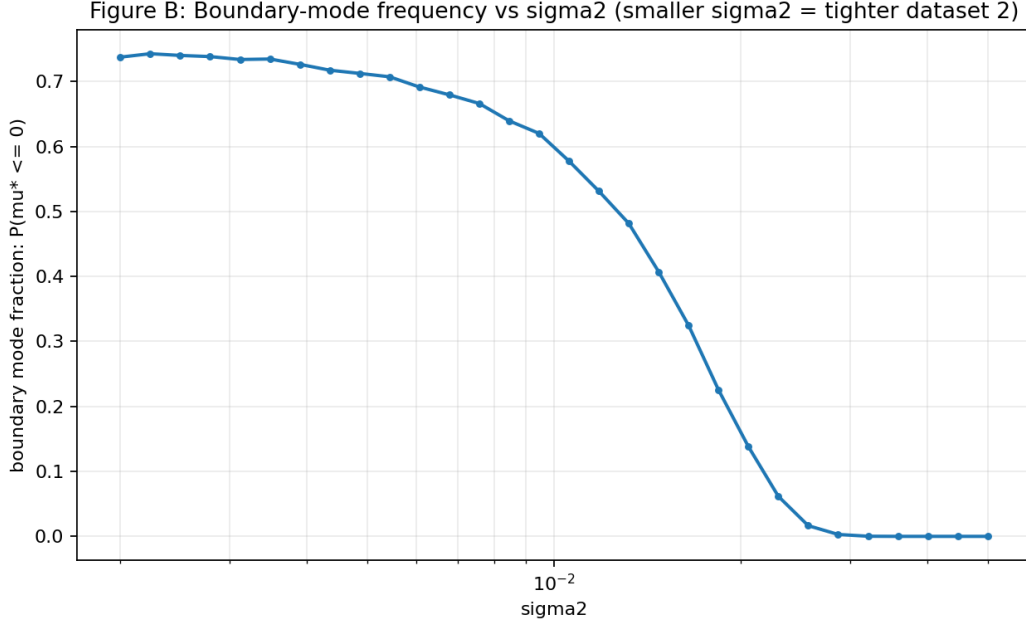


Figure 6: Toy sweep: as one constraint tightens, boundary-mode behavior (mode at $m = 0$ after truncation) becomes more frequent. This illustrates how tightening can amplify boundary sensitivity when packaged constraints are not mutually consistent.

$E = U \circ f$. Our audits operationalize the question “are these packages mutually predictive?” and the answer is: *not reliably*.

Three takeaways are most relevant for interpreting $\sum m_\nu$ shifts:

1. **Mismatch is large and directional.** The held-out mismatch statistics $\Delta\chi_{B|A}^2$ and $\Delta\chi_{A|B}^2$ are both substantial and asymmetric (Table 3), reinforcing that route mismatch is not captured by a single symmetric “tension” number.
2. **Mismatch is localized in TT high- ℓ within the overlap region.** The localization heatmaps (Figure 1) show that the dominant held-out failures arise from TT structure at high multipoles that both lenses cover. This supports treating the “curious shift” in [1] as entangled with packaging differences in how the two releases compress and model the same sky information.
3. **Common-staging does not resolve the mismatch.** Restricting D1 to the SPT2018 multipole support changes the profiled mismatch only at the few-percent level (Figure 2). This argues against a purely support/cut explanation and motivates reporting lens-swap audits alongside headline bounds when pipeline changes are in play.

Together with the reproduced direction of the DESI-driven $\sum m_\nu$ tightening under SPT2018→D1 (Section 7), these results support a conservative reading: $\sum m_\nu$ shifts can be *packaging-sensitive* whenever the packaged likelihood objects disagree in localized regions of the summary statistics.

9.2 Limitations

This note is deliberately scoped and has several limitations:

- **Completion/nuisance choices.** Our production $\sum m_\nu$ chains use a minimal nuisance sampling set for computational tractability, and our profiled audits optimize only a runtime-capped subset of nuisance parameters (Section 5.2). Broader nuisance freedom and alternative priors can shift tight sub-0.1 eV bounds.
- **Likelihood and baseline implementation details.** While we match the Planck-subset+PR4-lensing structure described in [1], small implementation choices (e.g. the low- ℓ EE variant available in a given likelihood wrapper) can propagate into tight derived bounds.
- **Finite compute and crude diagnostics.** Some of our MCMC diagnostics for $\sum m_\nu$ (ESS and split- $\hat{R}$ proxies [19, 20]) are necessarily crude given the expense of CMB+Boltzmann runs. Our aim is to reproduce qualitative behavior and provide audit localization, not to provide definitive converged limits beyond what is needed for the lens-swap comparison.
- **Audits over likelihood objects, not raw maps.** Our analyses operate on released likelihood objects. As such, the audits identify where the packaged summaries disagree, but they do not by themselves pinpoint whether the origin is calibration, beam/transfer functions, foreground modeling, mapmaking, or other pipeline components.

9.3 Next diagnostics (and non-novelty acknowledgement)

A decisive next step, emphasized already by [1], is to reprocess the 2018 raw data through the D1 pipeline. We explicitly *do not* claim novelty for proposing this idea. Our contribution is to clarify why such a test is well-motivated in packaging terms and to provide a set of quantitative artifacts that can be used to evaluate it.

Beyond reprocessing, SBT suggests two complementary diagnostic directions:

1. **Cross-lens posterior predictive checks (PPD-style audits) [24].** Rather than comparing only best-fit points, one can fit θ under lens A and forward-predict the summary statistics used by lens B (and vice versa), quantifying discrepancies as a function of spectrum and multipole. This would turn the localization maps here into a full predictive audit over posterior uncertainty.
2. **Minimal “rewrite mode” tests.** When mismatch localizes to specific blocks (e.g. TT high- ℓ), one can introduce a small surrogate correction (a template mode or calibration mode) whose role is to reduce cross-lens mismatch without being interpreted as new physics. Success would be measured by improved cross-lens predictivity and increased stability of $\sum m_\nu$ under staging and dataset routing; failure would sharpen the case that a deeper inconsistency (or missing modeling) remains.

Both directions are concrete and falsifiable and naturally complement standard parameter-tension narratives.

9.4 Practical recommendation

When pipeline/likelihood changes are known to move tight constraints, we recommend that headline parameter bounds (such as $\sum m_\nu$) be reported together with: (i) a directional cross-lens audit statistic (e.g. $\Delta\chi^2_{B|A}$ and $\Delta\chi^2_{A|B}$), (ii) a localization map by spectrum and multipole, and (iii) a common-staging control. These artifacts are small enough to be routine, yet they materially change how one should interpret shifts in packaged constraints under lens swaps.

10 Conclusion

We analyzed the “curious” $\sum m_\nu$ sensitivity to the SPT-3G 2018→D1 likelihood swap highlighted by [1] through the Six Birds / SBT lens [2, 3]. Treating likelihood releases as distinct lenses, we provided reproducible directional audits, localization by spectrum and multipole, and a common-staging control. The mismatch is large, directional, and dominated by TT high- ℓ structure inside the overlap region; restricting D1 to the SPT2018 multipole support does not resolve it. We also reproduced the direction of DESI-driven $\sum m_\nu$ tightening under the lens swap and clarified why boundary motion can arise generically under a hard physical gate.

Our central message is methodological: when a parameter bound moves under a lens swap, the appropriate response is to report (and interpret) the bound together with lens-swap audits and staging controls. Doing so turns an ambiguous “tension” into an actionable diagnosis and provides a disciplined path toward resolving packaging instabilities without prematurely attributing them to new cosmology.

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